MATH 350-2 Advanced Calculus

W.R. Casper

Department of Mathematics
California State University Fullerton

October 26, 2026

Outline

1. Real Analysis Lecture 17
   - Extreme Value Theorem
   - Continuity and Algebra
   - Connectedness
   - Uniform continuity

Outline

1. Real Analysis Lecture 17
   - Extreme Value Theorem
   - Continuity and Algebra
   - Connectedness
   - Uniform continuity

Extreme value theorem

Theorem

Suppose (S, d_S) is a metric space and $f : S \to \mathbb{R}$ is a function (to Euclidean space).
If f is continuous on a compact subset $A \subseteq S$, then there exists $a, b \in A$ such that

$$f(a) = \inf f(A), \quad \text{and} \quad f(b) = \sup f(A).$$

Extreme Value Theorem
Continuity and Algebra
Connectedness
Uniform continuity

Extreme value theorem

Extreme value theorem

Proof.

Extreme value theorem

Proof.

The set $f(A)$ is compact in $\mathbb{R}$, so it is closed and bounded.

Extreme value theorem

Proof.

The set $f(A)$ is compact in $\mathbb{R}$, so it is closed and bounded. The Completeness Axiom therefore tells us that it has an infimum $y = \inf(f(A))$ and supremum $z = \sup(f(A))$.

Extreme value theorem

Proof.

The set $f(A)$ is compact in $\mathbb{R}$, so it is closed and bounded. The Completeness Axiom therefore tells us that it has an infimum $y = \inf(f(A))$ and supremum $z = \sup(f(A))$. The infimum and supremum of a set are adherent points of that set.

Extreme value theorem

Proof.

The set $f(A)$ is compact in $\mathbb{R}$, so it is closed and bounded.
The Completeness Axiom therefore tells us that it has an infimum $y = \inf(f(A))$ and supremum $z = \sup(f(A))$.
The infimum and supremum of a set are adherent points of that set.
Therefore, since $f(A)$ is closed $y, z \in f(A)$.

Extreme value theorem

Proof.

The set $f(A)$ is compact in $\mathbb{R}$, so it is closed and bounded.
The Completeness Axiom therefore tells us that it has an infimum $y = \inf(f(A))$ and supremum $z = \sup(f(A))$.
The infimum and supremum of a set are adherent points of that set.
Therefore, since $f(A)$ is closed $y, z \in f(A)$.
This means that there exists $a, b \in A$ such that $y = f(a)$ and $z = f(b)$. □

Outline

1. Real Analysis Lecture 17
   - Extreme Value Theorem
   - Continuity and Algebra
   - Connectedness
   - Uniform continuity

Warm-up

Prove that the function $f : \mathbb{R}^2 \to \mathbb{R}$ defined by

$$f(x, y) = x + y$$

is continuous. (Everything Euclidean)

Warm-up

Prove that the function $f : \mathbb{R}^2 \to \mathbb{R}$ defined by

$$f(x, y) = xy$$

is continuous. (Everything Euclidean)

Warm-up

Prove that the function $f : \mathbb{R} \times (\mathbb{R}\backslash\{0\}) \to \mathbb{R}$ defined by

$$f(x, y) = x/y$$

is continuous. (Everything Euclidean)

Algebra and continuity

Theorem

Let (S, d) be a metric space and $f : S \to \mathbb{R}$ and $g : S \to \mathbb{R}$ be functions which are continuous at $a \in S$.

- *$f + g$ is continuous at a*
- *fg is continuous at a*
- *f/g is continuous at a as long as $g(a) \neq 0$*

Polynomials and Rational Functions

- constant functions are continuous
- $f(x) = x$ is continuous
- monomials x^n are continuous
- polynomials $a_0 + a_1 x + \cdots + a_n x^n$ are continuous
- rational functions $p(x)/q(x)$ are continuous on their domains

Outline

1. Real Analysis Lecture 17
   - Extreme Value Theorem
   - Continuity and Algebra
   - Connectedness
   - Uniform continuity

Characterization of intervals

Theorem

If $I \subseteq \mathbb{R}$ is an interval, then there does not exist two nonempty, disjoint open sets $U, V \subseteq \mathbb{R}$ (Euclidean space), with

$$I \subseteq U \cup V.$$

Connected spaces

Definition

A metric space (S, d) is called **disconnected** if there exists two nonempty, disjoint open subsets $U, V \subseteq S$ with $S = U \cup V$.

Connected spaces

Definition

A metric space (S, d) is called **disconnected** if there exists two nonempty, disjoint open subsets $U, V \subseteq S$ with $S = U \cup V$.
We call S **connected** if it is not disconnected.

Connected spaces

Definition

A metric space (S, d) is called **disconnected** if there exists two nonempty, disjoint open subsets $U, V \subseteq S$ with $S = U \cup V$.
We call S **connected** if it is not disconnected.
A subset $A \subseteq S$ is connected if the corresponding subspace (A, d) is connected.

Connected spaces

Definition

A metric space (S, d) is called **disconnected** if there exists two nonempty, disjoint open subsets $U, V \subseteq S$ with $S = U \cup V$.
We call S **connected** if it is not disconnected.
A subset $A \subseteq S$ is connected if the corresponding subspace (A, d) is connected.

Examples:

Connected spaces

Definition

A metric space (S, d) is called **disconnected** if there exists two nonempty, disjoint open subsets $U, V \subseteq S$ with $S = U \cup V$.
We call S **connected** if it is not disconnected.
A subset $A \subseteq S$ is connected if the corresponding subspace (A, d) is connected.

Examples:

- the real line with the Euclidean metric $(\mathbb{R}, d_{eucl})$ is connected

Connected spaces

Definition

A metric space (S, d) is called **disconnected** if there exists two nonempty, disjoint open subsets $U, V \subseteq S$ with $S = U \cup V$.
We call S **connected** if it is not disconnected.
A subset $A \subseteq S$ is connected if the corresponding subspace (A, d) is connected.

Examples:

- the real line with the Euclidean metric $(\mathbb{R}, d_{eucl})$ is connected
- the real line with the discrete metric $(\mathbb{R}, d_{disc})$ is disconnected

Connected spaces

Definition

A metric space (S, d) is called **disconnected** if there exists two nonempty, disjoint open subsets $U, V \subseteq S$ with $S = U \cup V$.
We call S **connected** if it is not disconnected.
A subset $A \subseteq S$ is connected if the corresponding subspace (A, d) is connected.

Examples:

- the real line with the Euclidean metric $(\mathbb{R}, d_{eucl})$ is connected
- the real line with the discrete metric $(\mathbb{R}, d_{disc})$ is disconnected
- euclidean space $\mathbb{R}^n$ is connected

Connected spaces

Definition

A metric space (S, d) is called **disconnected** if there exists two nonempty, disjoint open subsets $U, V \subseteq S$ with $S = U \cup V$.
We call S **connected** if it is not disconnected.
A subset $A \subseteq S$ is connected if the corresponding subspace (A, d) is connected.

Examples:

- the real line with the Euclidean metric $(\mathbb{R}, d_{eucl})$ is connected
- the real line with the discrete metric $(\mathbb{R}, d_{disc})$ is disconnected
- euclidean space $\mathbb{R}^n$ is connected
- an interval $I \subseteq \mathbb{R}$ with the Euclidean metric is connected

Challenge!

Problem

Prove that the set $\mathbb{Q} \subseteq \mathbb{R}$ is disconnected.

Real line is connected

Theorem

The real line $\mathbb{R}$ is connected.

Real line is connected

Theorem

The real line $\mathbb{R}$ is connected.

Proof.

Real line is connected

Theorem

The real line $\mathbb{R}$ is connected.

Proof.

Suppose that $\mathbb{R}$ is disconnected.

Real line is connected

Theorem

The real line $\mathbb{R}$ is connected.

Proof.

Suppose that $\mathbb{R}$ is disconnected.
Then $\mathbb{R} = U \cup V$ where:

Real line is connected

Theorem

The real line $\mathbb{R}$ is connected.

Proof.

Suppose that $\mathbb{R}$ is disconnected.
Then $\mathbb{R} = U \cup V$ where:

- $U \neq \varnothing$, $V \neq \varnothing$

Real line is connected

Theorem

The real line $\mathbb{R}$ is connected.

Proof.

Suppose that $\mathbb{R}$ is disconnected.
Then $\mathbb{R} = U \cup V$ where:

- $U \neq \varnothing$, $V \neq \varnothing$
- $U \cap V = \varnothing$

Real line is connected

Theorem

The real line $\mathbb{R}$ is connected.

Proof.

Suppose that $\mathbb{R}$ is disconnected.
Then $\mathbb{R} = U \cup V$ where:

- $U \neq \varnothing$, $V \neq \varnothing$
- $U \cap V = \varnothing$
- U, V are open

Real line is connected

Theorem

The real line $\mathbb{R}$ is connected.

Proof.

Suppose that $\mathbb{R}$ is disconnected.
Then $\mathbb{R} = U \cup V$ where:

- $U \neq \varnothing$, $V \neq \varnothing$
- $U \cap V = \varnothing$
- U, V are open
- $U \cup V = \mathbb{R}$

Real line is connected

Theorem

The real line $\mathbb{R}$ is connected.

Proof.

Suppose that $\mathbb{R}$ is disconnected.
Then $\mathbb{R} = U \cup V$ where:

- $U \neq \varnothing$, $V \neq \varnothing$
- $U \cap V = \varnothing$
- U, V are open
- $U \cup V = \mathbb{R}$

In particular, U is a nonempty, proper subset of $\mathbb{R}$ which is both open and closed. □

Real line is connected

Theorem

The real line $\mathbb{R}$ is connected.

Real line is connected

Theorem

The real line $\mathbb{R}$ is connected.

Proof.

Real line is connected

Theorem

The real line $\mathbb{R}$ is connected.

Proof.

By the Representation Theorem for Open Sets:

Real line is connected

Theorem

The real line $\mathbb{R}$ is connected.

Proof.

By the Representation Theorem for Open Sets:

$$U = (a_1, b_1) \cup (a_2, b_2) \cup \dots$$

Real line is connected

Theorem

The real line $\mathbb{R}$ is connected.

Proof.

By the Representation Theorem for Open Sets:

$$U = (a_1, b_1) \cup (a_2, b_2) \cup \dots$$

is a union of disjoint open intervals.

Real line is connected

Theorem

The real line $\mathbb{R}$ is connected.

Proof.

By the Representation Theorem for Open Sets:

$$U = (a_1, b_1) \cup (a_2, b_2) \cup \dots$$

is a union of disjoint open intervals.
In particular, b_1 is an accumulation point of U but not an element of U.

Real line is connected

Theorem

The real line $\mathbb{R}$ is connected.

Proof.

By the Representation Theorem for Open Sets:

$$U = (a_1, b_1) \cup (a_2, b_2) \cup \dots$$

is a union of disjoint open intervals.
In particular, b_1 is an accumulation point of U but not an element of U.
This contradicts the fact that U is closed. □

Images of connected sets

Theorem

Let (S, d_S) and (T, d_T) be metric spaces and $f : S \to T$ be a function.
If f is continuous and S is connected, then $f(S)$ is connected.

Images of connected sets

Theorem

Let (S, d_S) and (T, d_T) be metric spaces and $f : S \to T$ be a function.
If f is continuous and S is connected, then $f(S)$ is connected.

Proof.

Images of connected sets

Theorem

Let (S, d_S) and (T, d_T) be metric spaces and $f : S \to T$ be a function.
If f is continuous and S is connected, then $f(S)$ is connected.

Proof.

Suppose $f(S)$ is disconnected.

Images of connected sets

Theorem

Let (S, d_S) and (T, d_T) be metric spaces and $f : S \to T$ be a function.
If f is continuous and S is connected, then $f(S)$ is connected.

Proof.

Suppose $f(S)$ is disconnected.
Then $f(S) = U \cup V$ for nonempty, disjoint, open sets $U, V \subseteq f(S)$.

Images of connected sets

Theorem

Let (S, d_S) and (T, d_T) be metric spaces and $f : S \to T$ be a function.
If f is continuous and S is connected, then $f(S)$ is connected.

Proof.

Suppose $f(S)$ is disconnected.
Then $f(S) = U \cup V$ for nonempty, disjoint, open sets $U, V \subseteq f(S)$.
Then $f^{-1}(U)$ and $f^{-1}(V)$ are non-empty, disjoint, open sets and

$$f^{-1}(U) \cup f^{-1}(V) = f^{-1}(U \cup V) = f^{-1}(f(S)) = S.$$

Images of connected sets

Theorem

Let (S, d_S) and (T, d_T) be metric spaces and $f : S \to T$ be a function.
If f is continuous and S is connected, then $f(S)$ is connected.

Proof.

Suppose $f(S)$ is disconnected.
Then $f(S) = U \cup V$ for nonempty, disjoint, open sets $U, V \subseteq f(S)$.
Then $f^{-1}(U)$ and $f^{-1}(V)$ are non-empty, disjoint, open sets and

$$f^{-1}(U) \cup f^{-1}(V) = f^{-1}(U \cup V) = f^{-1}(f(S)) = S.$$

This is a contadiction. □

Any interval is connected

Theorem

Any open interval is connected

Any interval is connected

Theorem

Any open interval is connected

Proof.

Any interval is connected

Theorem

Any open interval is connected

Proof.

We can make any interval as a continuous image:

$$f : \mathbb{R} \to \mathbb{R}$$

$$f(x) = \frac{b+a}{2} + \frac{(b-a)x}{1+x^2}$$

Any interval is connected

Theorem

Any open interval is connected

Proof.

We can make any interval as a continuous image:

$$f : \mathbb{R} \to \mathbb{R}$$

$$f(x) = \frac{b+a}{2} + \frac{(b-a)x}{1+x^2}$$

$$f(\mathbb{R}) = (a, b), \quad a < b$$



Any interval is connected

Theorem

Any closed interval is connected

Any interval is connected

Theorem

Any closed interval is connected

Proof.

We can make any interval as a continuous image:

$$f : \mathbb{R} \to \mathbb{R}$$

$$f(x) = a + \frac{2(b-a)x^2}{1+x^4}$$

Any interval is connected

Theorem

Any closed interval is connected

Proof.

We can make any interval as a continuous image:

$$f : \mathbb{R} \to \mathbb{R}$$

$$f(x) = a + \frac{2(b-a)x^2}{1+x^4}$$

$$f(\mathbb{R}) = [a, b], \quad a < b$$



Any interval is connected

Theorem

Any half-open interval is connected

Any interval is connected

Theorem

Any half-open interval is connected

Proof.

Any interval is connected

Theorem

Any half-open interval is connected

Proof.

We can make any interval as a continuous image:

$$f : \mathbb{R} \to \mathbb{R}$$

$$f(x) = a + \frac{(b-a)x^2}{1+x^2}$$

Any interval is connected

Theorem

Any half-open interval is connected

Proof.

We can make any interval as a continuous image:

$$f : \mathbb{R} \to \mathbb{R}$$

$$f(x) = a + \frac{(b-a)x^2}{1+x^2}$$

$$f(\mathbb{R}) = [a, b), \quad a < b$$

Any interval is connected

Theorem

Any half-open interval is connected

Proof.

We can make any interval as a continuous image:

$$f : \mathbb{R} \to \mathbb{R}$$

$$f(x) = a + \frac{(b-a)x^2}{1+x^2}$$

$$f(\mathbb{R}) = [a, b), \quad a < b$$

$$f(\mathbb{R}) = (b, a], \quad a > b$$

Intermediate value theorem

Theorem (Intermediate Value Theorem)

Suppose $f : [a, b] \to \mathbb{R}$ *is a continuous function. Then for any value* y *between* $f(a)$ *and* $f(b)$*, there exists* $c \in [a, b]$ *with* $f(c) = y$*.*

Intermediate value theorem

Proof.

Intermediate value theorem

Proof.

Let $J = f([a, b])$.

Intermediate value theorem

Proof.

Let $J = f([a, b])$.
Then J is connected!

Intermediate value theorem

Proof.

Let $J = f([a, b])$.
Then J is connected!
Let y be between $f(a)$ and $f(b)$, and suppose that $y \notin J$.

Intermediate value theorem

Proof.

Let $J = f([a, b])$.
Then J is connected!
Let y be between $f(a)$ and $f(b)$, and suppose that $y \notin J$.
Without loss of generality, assume $f(a) < f(b)$.

Intermediate value theorem

Proof.

Let $J = f([a, b])$.
Then J is connected!
Let y be between $f(a)$ and $f(b)$, and suppose that $y \notin J$.
Without loss of generality, assume $f(a) < f(b)$.
Then $J = U \cup V$ for

$$U = (-\infty, y) \cap J \quad \text{and} \quad V = (y, \infty) \cap J$$

Intermediate value theorem

Proof.

Let $J = f([a, b])$.
Then J is connected!
Let y be between $f(a)$ and $f(b)$, and suppose that $y \notin J$.
Without loss of generality, assume $f(a) < f(b)$.
Then $J = U \cup V$ for

$$U = (-\infty, y) \cap J \quad \text{and} \quad V = (y, \infty) \cap J$$

These are nonempty, disjoint, and open in the subspace J of $\mathbb{R}$.

Intermediate value theorem

Proof.

Let $J = f([a, b])$.
Then J is connected!
Let y be between $f(a)$ and $f(b)$, and suppose that $y \notin J$.
Without loss of generality, assume $f(a) < f(b)$.
Then $J = U \cup V$ for

$$U = (-\infty, y) \cap J \quad \text{and} \quad V = (y, \infty) \cap J$$

These are nonempty, disjoint, and open in the subspace J of $\mathbb{R}$.
This is a contradiction! □

Outline

1. Real Analysis Lecture 17
   - Extreme Value Theorem
   - Continuity and Algebra
   - Connectedness
   - Uniform continuity

Uniform continuity

Definition

Let (S, d_S) and (T, d_T) be metric spaces and $A \subseteq S$. We say that a function $f : A \to T$ is **uniformly continuous** on A if for all $\epsilon > 0$ there exists $\delta > 0$ such that

$$d_S(x, y) < \delta \Rightarrow d_T(f(x), f(y)) < \epsilon$$

for all $x, y \in A$.

- f continuous on A means f is continuous at every $a \in A$
- δ can depend on ϵ and a (not on x)
- with uniform continuity, δ can't depend on a ...

Examples

- x is uniformly continuous on $\mathbb{R}$
- x^2 is NOT uniformly continuous on $\mathbb{R}$
- $1/x$ is continuous on $(0, 1)$
- $1/x$ is NOT uniformly continuous on $(0, 1)$

Relationship with compactness

Theorem (Heine)

Let (S, d_S) and (T, d_T) be metric spaces and $A \subseteq S$ be compact. Suppose that $f : A \to T$ is continuous on A, then f is uniformly continuous on A.